

Q: 2(i)

A: 2(i)

A polypeptide folds through 4 levels of protein structure :-

① Primary structure :-

The specific sequence of amino acids joined by peptide bonds.

② Secondary structure :-

The chain forms α -helices and β -pleated sheets, mainly due to hydrogen bonding b/w peptide bonds.

③ Tertiary structure :-

The secondary structures fold further into a specific 3D shape, held by interaction b/w amino acids chains, such as hydrogen bonds, ionic bonds and disulphide bonds.

④ Quaternary structure :-

Two or more folded polypeptide chains may join together to form a functional protein.

Q:2(ii)

A:2(ii)

malaria is difficult to eradicate because of several biological factors involving Plasmodium and its mosquito vector.

① Complex life cycle - Plasmodium has different stages in humans and mosquitoes, making it difficult to target all stages with one treatment.

② Mosquitoes reproduce rapidly:-
Large number of mosquitoes can breed, making vector control difficult.

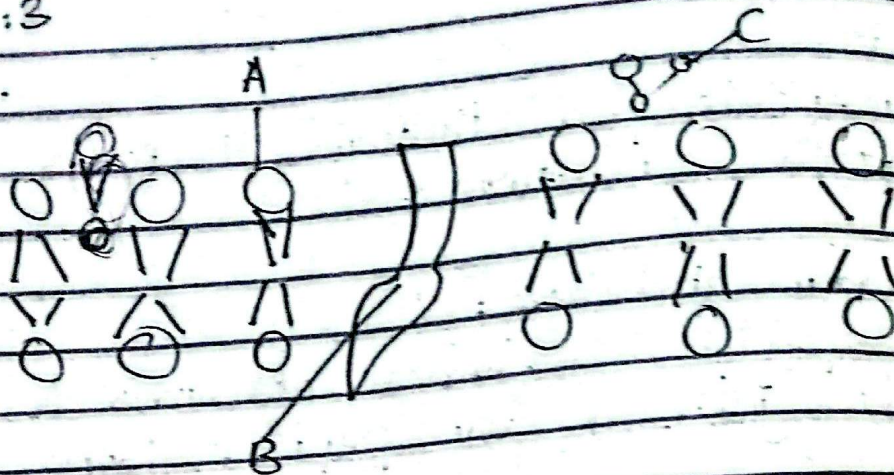
~~Insecticide resistance:-~~
③ Drug resistance:-
Mosquitoes populations can develop resistance to insecticides used to control them.

④ Drug resistance:-
Plasmodium can develop resistance to antimalarial drugs, reducing their effectiveness.

Q:3

A:3

a.



- A. hydrophilic head of a phospholipid
- B. Integral protein
- C. Carbohydrate chain.

b. Cell membrane is a thin sheet like covering of the cell. ~~Chemically~~ Chemically it is composed of proteins 60-80% & phospholipids 20-40% and traces of carbohydrates.

c. The fluid mosaic model postulates that the cell membrane consists of double layer of phospholipids in which proteins are incorporated in a mosaic fashion. The proteins can move laterally within the phospholipid bilayer making the membrane fluid.

MCQs:-

1. B

because 7/100 are the malaria patients, 100/107 and 50/100 doesn't make any sense.

2. D

hypothesis can be true or false, deduction is using logical terms like if or then. A good scientific theory is giving theory after experiment and explains how and why it happens.

3. A

bio means biology and the combined term is informatics.

4. C

Specifics comes only in small letter while genes comes only in capital letter.

5. A

6. D

The other options are linked to cytoskeleton.

7. B

8. C

9. A

10. D

11. C

12. B

13. D

14. B

15. A

16. C

17. D

18. D

19. C

20. B